Rebuttal-Related Supplementary Additions

This document is prepared to provide the reviewers with additional details and clarifications related to our rebuttal. We sincerely appreciate the reviewers' careful reading and constructive feedback, and hope the following supplementary materials help explain the relevant points more clearly.

1 Natural Degradation Evidence in Source Videos

Because the source videos are collected from in-the-wild datasets, part of MIME already contains realistic natural degradation before any controlled corruption is applied. Representative natural degradation types include low-light concealment, side-facing or turning-away heads, foreground/object occlusion, and hand/body occlusion. These naturally imperfect cases are important because they better reflect real deployment scenarios than purely clean source videos.

Figure 1 shows four representative examples. The caption format follows *Subset + Emotion + Degradation Type*.

Subset 3 (FSM), Neutral
low-light / low visibility

Subset 2 (FDM), Fear
side-facing / turning-away head

Subset 3 (FSM), Fear
foreground / object occlusion

Subset 2 (FDM), Anger
hand / body occlusion

Figure 1: Representative natural degradation evidence already present in source videos before any controlled corruption is applied.

2 Updated Metric Formulation

To make the missing-aware scoring explicit, the revised manuscript updates Eqs. (1) and (2) as follows:

$$\mathrm{CPS} = \frac{\sum_{m\in\mathcal{M}} I_m \cdot s_m^{\mathrm{CPS}}}{\sum_{m\in\mathcal{M}} I_m} \tag{1}$$

$$\mathrm{AIS} = \frac{\sum_{m\in\mathcal{M}} I_m \cdot s_m^{\mathrm{AIS}}}{\sum_{m\in\mathcal{M}} I_m} \tag{2}$$

where $\mathcal{M} = \{\text{face}, \text{body}, \text{scene}, \text{audio}\}$ and $I_m \in \{0, 1\}$ indicates whether modality m is valid for scoring. Specifically, $I_m = 1$ if modality m is available or degraded-but-observable, and $I_m = 0$ if modality m is truly missing.

3 Additional Rebuttal-Stage Experimental Results

3.1 Cross-Judge Validation on a Stratified Subset

We replaced the original Qwen-family judge with `Gemini-3.1-flash-lite` and re-evaluated a stratified subset covering all 7 subsets and all 7 emotion categories (5 samples per cell; 245 samples in total). Rankings are compared by HCI. Relative to the original Qwen-based ranking, the top-5 models remain exactly the same, and only the last two models switch order, yielding a Spearman rank correlation of 0.9643.

Table 1: Cross-judge validation on the 245-sample stratified subset using `Gemini-3.1-flash-lite` as an independent judge. The top three results in each metric column are highlighted in red (1^{st}), blue (2^{nd}), and yellow (3^{rd}).

Model	CPS	AIS	LCS	HCI
Qwen3.5-plus	**7.24**	**5.01**	4.29	**5.68**
Gemini-3.1-flash-lite-preview	6.96	4.84	4.59	5.61
Qwen3.5-27b	7.17	**5.01**	3.92	5.54
Doubao-seed-2-0-lite	6.80	4.67	**4.65**	5.52
GPT-5-mini	6.54	4.54	3.22	4.95
GPT-5.4-nano	5.92	4.06	2.82	4.43
GPT-5-nano	5.93	3.85	2.69	4.34

3.2 Prompt Sensitivity Under the Gemini Judge

We further tested three prompt variants for the Gemini judge: the original prompt (P1), a shorter and more standardized reformulation (P2), and a judge-neutrality-aware prompt (P3). Rankings are again compared by HCI. The relative model ranking remains stable: $\text{Spearman}(\text{P1}, \text{P2}) = 1.0000$ and $\text{Spearman}(\text{P1}, \text{P3}) = 0.9286$.

Table 2: Prompt sensitivity under the Gemini judge on the same stratified subset, reported using HCI. The top three results in each prompt column are highlighted in red (1^{st}), blue (2^{nd}), and yellow (3^{rd}).

Model	P1	P2	P3
Qwen3.5-plus	**5.68**	**5.98**	**5.97**
Gemini-3.1-flash-lite-preview	5.61	5.95	5.92
Qwen3.5-27b	5.54	5.81	5.73
Doubao-seed-2-0-lite	5.52	5.75	5.89
GPT-5-mini	4.95	5.43	5.43
GPT-5.4-nano	4.43	4.86	4.86
GPT-5-nano	4.34	4.78	4.89

3.3 HCI Sensitivity to Alternative Weight Settings

We also tested whether the overall ranking is sensitive to the empirical HCI weighting in Eq. (3). Table 3 reports the HCI scores under four weight settings. The top three results in each column are highlighted in red (1^{st}), blue (2^{nd}), and yellow (3^{rd}). The resulting rank correlations remain high, ranging from 0.93 to 1.00.

Table 3: HCI sensitivity under different (α, β) settings. Spearman correlation is computed against the default ranking under $(\alpha, \beta) = (0.4, 0.3)$.

Model	Default (0.4,0.3)	S1 (0.4,0.2)	S2 (0.3,0.4)	S3 (0.2,0.5)
Qwen3.5-plus	**5.605**	5.516	**5.467**	5.329
Gemini-3.1-flash	5.592	**5.536**	5.462	**5.332**
Qwen3.5-27b	5.463	5.382	5.319	5.175
Doubao-seed-2-0	5.455	5.424	5.308	5.161
GPT-5-mini	4.734	4.640	4.585	4.436
gpt-5-nano	4.315	4.204	4.168	4.021
GPT-5.4-nano	4.234	4.130	4.090	3.946
Spearman (vs. Default)	1.00	0.93	1.00	0.96

4 Additional Notes on Subset Construction

Through subject tracking and face-region localization, we establish a high-quality full-modality baseline and then construct seven subsets. Importantly, when a source clip already contains sufficiently strong natural degradation—such as a back-facing head, severe side-facing pose, partial natural occlusion, low-light concealment, or unstable face visibility—we do not add extra artificial blur on top of that naturally degraded evidence. By contrast, when the source clip does not contain such naturally strong face degradation, we apply controlled Gaussian blur inside the facial mask to simulate the intended incomplete-modality condition.

This design keeps MIME closer to realistic in-the-wild imperfection while preserving a systematic subset taxonomy. As a result, the benchmark should be understood as a combination of *natural incompleteness already present in the source videos* and *controlled synthetic degradation applied when needed to complete the target subset design.*

- **Subset 1 (MIME-FM):** Full modality. Intact audio-visual clips are kept as the baseline, although some source clips may still contain natural imperfections such as mild occlusion or pose variation.

- **Subset 2 (MIME-FDM):** Face details missing. For clips without sufficiently strong natural facial-detail loss, we apply light Gaussian blur within the facial mask so that subtle details are weakened while the coarse facial layout remains usable.

- **Subset 3 (MIME-FSM):** Face structures missing. For clips without sufficiently strong natural structural loss, we apply heavier Gaussian blur so that the face is no longer reliable for emotion inference.

- **Subset 4 (MIME-VMM):** Visual modality missing. All visual frames are removed, leaving an audio-only condition.

- **Subset 5 (MIME-FDAM):** Face details and audio modality missing. This combines the FDM-style face degradation with removed audio.

- **Subset 6 (MIME-FSAM):** Face structures and audio modality missing. This combines the FSM-style face degradation with removed audio.

- **Subset 7 (MIME-AMM):** Audio modality missing. The video remains while the audio track is removed.

5 Chain-of-Emotion (CoE) Generation

To endow our benchmark with rich, fine-grained reasoning annotations, we design a comprehensive four-stage pipeline utilizing Qwen3-Omni to generate high-quality Chain-of-Emotion. This pipeline explicitly guides models to decouple and articulate their observations across individual modalities prior to predicting the final emotion.

Stage I: Foundational Cue Extraction. The model is prompted to systematically decouple the input into distinct fine-grained modalities: Face, Body, Scene, and Audio. For uncorrupted or inherently unimodal subsets (Subsets 1 and 4), this extraction operates directly on the target input. However, for visually degraded Subsets (Subsets 2, 3, 5, and 6), we first input the original clear video to extract a comprehensive, factual baseline of objective physical elements.

Stage II: Conditional Remained-Cue Verification. This stage is exclusively triggered for subsets involving facial blurring (Subsets 2, 3, 5, and 6). We input the degraded video alongside the baseline cues extracted in Stage I. The model is instructed to rigorously cross-reference these inputs and explicitly delete any facial details or structural information that are no longer discernible due to the applied Gaussian blur. This critical step dynamically prunes the cue set and suppresses multi-modal hallucinations.

Stage III: Ground-truth Consistency Filtering. To guarantee the validity and rigorousness of the benchmark, we implement a strict programmatic validation step. The predicted emotion derived from the verified cue set is cross-checked against the original ground-truth label. Samples where even state-of-the-art models cannot deduce the correct label are discarded, ensuring that the remaining challenging subsets in MIME are still fundamentally solvable.

Stage IV: Structured CoE Generation. The final verified samples are processed through a structured prompt, where the MLLM acts as a professional expert in psychological and micro-expression analysis, to generate the formalized Chain-of-Emotion (CoE) in a JSON format. We mandate a strict tripartite logic flow:

- **Scene Understanding:** Characterizing the objective multimodal cues present in the input, extracting factual information exclusively from the available modalities.
- **Emotional Analysis:** Executing cognitive reasoning to map the previously identified cues to internal emotional states.
- **Predicted Emotion:** Deriving the final discrete emotion category, which strictly aligns with the ground-truth label, based on the preceding systematic analysis.

Algorithm 1 Chain-of-Emotion (CoE) Generation Pipeline

Require: Target Video $\mathcal{V}$ (clear or degraded), Ground Truth Emotion $\mathcal{E}$, Subset Type $\mathcal{S} \in \{1, 2, \ldots, 7\}$
Ensure: Structured CoE Output or Rejected Status

```
    // Stage I: Foundational Cue Extraction
if S == 1 or S == 4 then
    Oracle_Cues ← MLLM(Prompt_Stage1, V, E)
else
    V_clear ← Get_Clear_Version(V)
    Oracle_Cues ← MLLM(Prompt_Stage1, V_clear, E)
end if
    // Stage II: Conditional Remained-Cue Verification
if S ∈ {2, 3, 5, 6} then
    Blur_Level ← Get_Blur_Level(V)
    if Blur_Level == Light then
        Verified_Cues ← MLLM(Prompt_Stage2_Light, V, Oracle_Cues)
    else
        Verified_Cues ← MLLM(Prompt_Stage2_Heavy, V, Oracle_Cues)
    end if
else
    Verified_Cues ← Oracle_Cues
end if
    // Stage III: Ground-truth Consistency Filtering
Ê ← Infer_Emotion(Verified_Cues)
if Ê ≠ E then
    return Rejected
end if
    // Stage IV: Structured CoE Generation
CoE_Output ← MLLM(Prompt_Stage4, V, Verified_Cues, E)
return CoE_Output
```

Example of the 4-Stage CoE Generation Pipeline

To concretely illustrate our proposed data construction methodology, we provide a complete, step-by-step pipeline execution for a heavily degraded sample (Subset 3: Face Structures Missing). This example explicitly demonstrates the complete prompts and corresponding outputs at each stage, showcasing how the system extracts foundational cues, rigorously filters out hallucinated facial features, performs blind emotion inference, and synthesizes the structured Chain-of-Emotion.

Pipeline Execution Record | **Subset:** Subset 3 (FSM) | **GT Emotion:** Anger

STAGE I: Foundational Cue Extraction

Input (System Prompt + Original Clear Video):

Role: Multimodal Emotion Analysis Expert
Task: Analyze the provided **CLEAR, UNBLURRED video**. The ground truth emotion is **Anger**. Your goal is to extract a structured "Visual & Audio Evidence List" based on the raw footage. **Crucial:** Focus specifically on extracting and detailing cues that constitute the evidence for **Anger**.
ANONYMITY & OBJECTIVITY PROTOCOL
* **IGNORE EXTERNAL KNOWLEDGE:** Do NOT use specific actor names or set names.
* **USE GENERIC LABELS:** Use gender-specific labels like "The man in [clothing]".
* **EXCEPTION:** Only write a name if it is explicitly spoken in the audio **Caption**.
# **Extraction Guidelines**
1. **SCENE & CONTEXT:** Describe background, social setting, and specific physical interactions. **Target Identity:** You MUST explicitly describe the Target Person's appearance. Priority Rule: 1. Clearest Face. 2. Largest Body Area.
2. **BODY & HEAD:** Describe specific hand gestures, body posture, head orientation, and movement speed.
3. **AUDIO & SPEECH ANALYSIS:** Step 1: Identify the Speaker. Category A: [Target Person Speaking]. Verify this matches Target Identity. Output Caption and Tone (Pitch, Volume, Speed). Category B: [Other Person Speaking]. Output Caption ONLY. Category C: [Scene Noise]. Category D: [Quiet/No Voice].
4. **FACIAL MOVEMENTS:** Analyze the face using **Facial Action Units (AU)** concepts for accuracy, but **OUTPUT ONLY natural language descriptions**. **FORBIDDEN:** Do NOT write "AU4". Describe specific organ movement clearly.
Output Constraint DO NOT write an introduction. **Output strictly** in the specified section format.

Output (Oracle Cues):

- **### SECTION A: SCENE & CONTEXT**
  - The scene takes place indoors, likely in a residential hallway or entryway, with a staircase railing visible in the background and a decorative light fixture overhead.

- The woman is standing and facing the man, engaged in a direct conversation. Another person in a red hat walks past them in the background, indicating a casual, possibly domestic setting.

- **### SECTION B: BODY & HEAD**

  - The woman is standing upright, facing the man, with her body oriented toward him, indicating direct engagement... The man is seen from the side and back, with his head turned toward the woman... There is no visible aggressive physical interaction.

- **### SECTION C: AUDIO ANALYSIS**

  - **Source Category:** Target Person (The woman in the white floral dress)

  - **Caption:** "I didn't lie. You made an assumption I didn't contradict it."

  - **Tonal/Sound Details:** The woman's voice is firm and slightly raised in pitch, indicating defensiveness and irritation. Her speech is delivered at a moderate pace, with a clear, assertive tone...

- **### SECTION D: FACIAL MOVEMENTS**

  - Brows lowered and slightly furrowed, indicating frustration or annoyance.

  - Eyes wide and fixed on the man... Mouth slightly open and lips pressed together...

STAGE II: Conditional Remained-Cue Verification

Input (System Prompt + Degraded Video + Oracle Cues):

\# **Role:** Privacy-Preserving Video Analyst (Heavy Blur Specialist)
\# **Context:** You are analyzing a video with **LEVEL 3 (HEAVY) BLUR** on the face. Blur Status: The face is a featureless blob/pixelated block. **Goal: STRICTLY REJECT** all facial details to prevent Hallucination.
\# **NEGATIVE CONSTRAINT (CRITICAL)**
* **NO EMOTION WORDS:** You are an objective observer. **DO NOT** use emotion adjectives like "angry", "happy" in your output.
* **USE PHYSICAL DESCRIPTIONS:** Instead of "angry tone", use "loud, fast-paced tone".
\# **Protocol**
1. **Identity Extraction:** Create standalone fact line: **[REMAINED] (Identity) {Identity}**.
2. **Audio Verification:** Validate the Source Category.
3. **Body/Scene Verification:** Mostly keep them. Prefix with **[REMAINED]**.
4. **Face Verification:** The face is considered INVISIBLE. **DELETE** all specific facial cues.

Output (Verified Facts):

- **[REMAINED] (Identity)** The man in the purple shirt with colorful embroidery
- **[REMAINED] (Scene)** The scene takes place indoors, likely in a residential hallway or entryway, with a staircase railing visible in the background and a decorative light fixture overhead.
- **[REMAINED] (Body)** The woman is standing upright, facing the man, with her body oriented toward him, indicating direct engagement... The man is seen from the

side and back, with his head turned toward the woman... There is no visible aggressive physical interaction.

- **[REMAINED] (Audio)** [Source: Target Person] "I didn't lie. You made an assumption I didn't contradict it." - Tone: loud, fast-paced, assertive.
- **[DELETED] (Face)** *All facial cues from Stage I are strictly removed.*

STAGE III: Ground-truth Consistency Filtering

Input (System Prompt + Verified Facts):

\# **Role** Blind Emotion Inference Engine
\# **Task** Based **SOLELY** on the verified facts above, infer the most likely emotion.
\# **Inference Logic** 1. Semantic Analysis (Caption content). 2. Tonal Analysis (Primary). 3. Contextual Analysis (Scene interactions and Body Language).
\# **Output Format** JSON Only.

Output (Filter Result):


```
{
  "predicted_emotion": "Anger",
  "confidence": "High",
  "emotion_cues": "Target shouted loudly and fast-paced, asserting 'I
     didn't lie. You made an assumption I didn't contradict it.'",
  "reasoning": "The target person's voice is loud and fast-paced,
     indicating assertiveness often associated with anger. The content
     of the speech -- denying a lie while accusing the other party of
     making an assumption -- suggests defensive frustration. The body
     language shows direct engagement, with both individuals facing each
     other, which reinforces the intensity of the interaction. No visual
     or auditory cues suggest laughter, surprise, or sadness. The
     combination of tone and semantic content strongly supports Anger as
     the dominant emotion."
}
```


Note: Since Predicted Emotion matches Ground Truth (Anger), the sample proceeds.

STAGE IV: Structured CoE Generation

Input (System Prompt + Verified Facts + Inferred Cues):

Role Multimodal CoT Generator (Visual-Grounded)
Goal Write a logical analysis (Chain-of-Emotion) that explains **WHY** the video expresses **Anger**, based on the verified facts and emotion cues.
Output Rules 1. Integration over Listing. 2. Strict Fact Adherence. **DO NOT** hallucinate facial cues if they are not listed in the facts (especially for Level 2/3).
Format 1. Scene Understanding (MANDATORY: Start by identifying the protagonist. IF Level == 2 or 3: You MUST write: "The face is heavily blurred.") 2. Emotional Analysis (Analyze in order: Facial Premise, Audio Caption & Tone, Scene Interaction, Body Language). 3. Conclusion.

Output (Final CoE Annotation):

1. Scene Understanding

The man in the purple shirt with colorful embroidery stands in a residential hallway, facing a woman. The face is heavily blurred. A staircase railing and decorative light fixture are visible, indicating an indoor setting. Both individuals remain upright, engaged in direct conversation.

2. Emotional Analysis

Although the face is heavily blurred, we can still infer anger from the following cues. The man in the purple shirt says "I didn't lie. You made an assumption I didn't contradict it." with a loud, fast-paced, assertive tone. The scene shows no aggressive physical interaction, yet their upright, face-to-face stance suggests confrontation.

3. Conclusion

Anger